

NLP Libraries

NLTK

spaCy

re

Gensim

FastText

Libraries Used in the NLP Course

This Chapter organizes the libraries into NLP-specific tools and general-purpose Python libraries.

NLP-focused libraries

NLTK
spaCy
re (Regular Expressions)
Gensim
FastText

General-purpose Python libraries

NumPy
pandas
scikit-learn
Matplotlib

NLTK — Natural Language Toolkit

NLTK is one of the major libraries for Natural Language Processing.

The acronym stands for Natural Language Toolkit.

the project was created in 2001 by Steven Bird, Edward Loper, and Ewan Klein.

version 3.10.3 as the latest version at the time of the course material (Oct 2026).

Installation command:

```
pip install nltk
```



Verifying and Downloading NLTK Resources

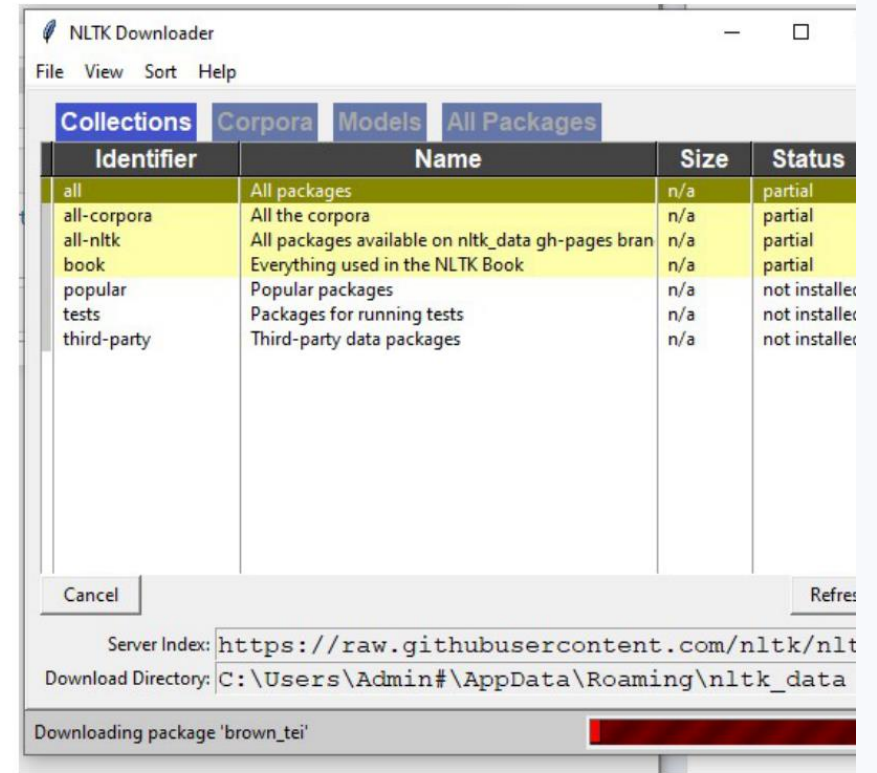
Check the installed version

```
import nltk
nltk.__version__
```

Open the NLTK downloader

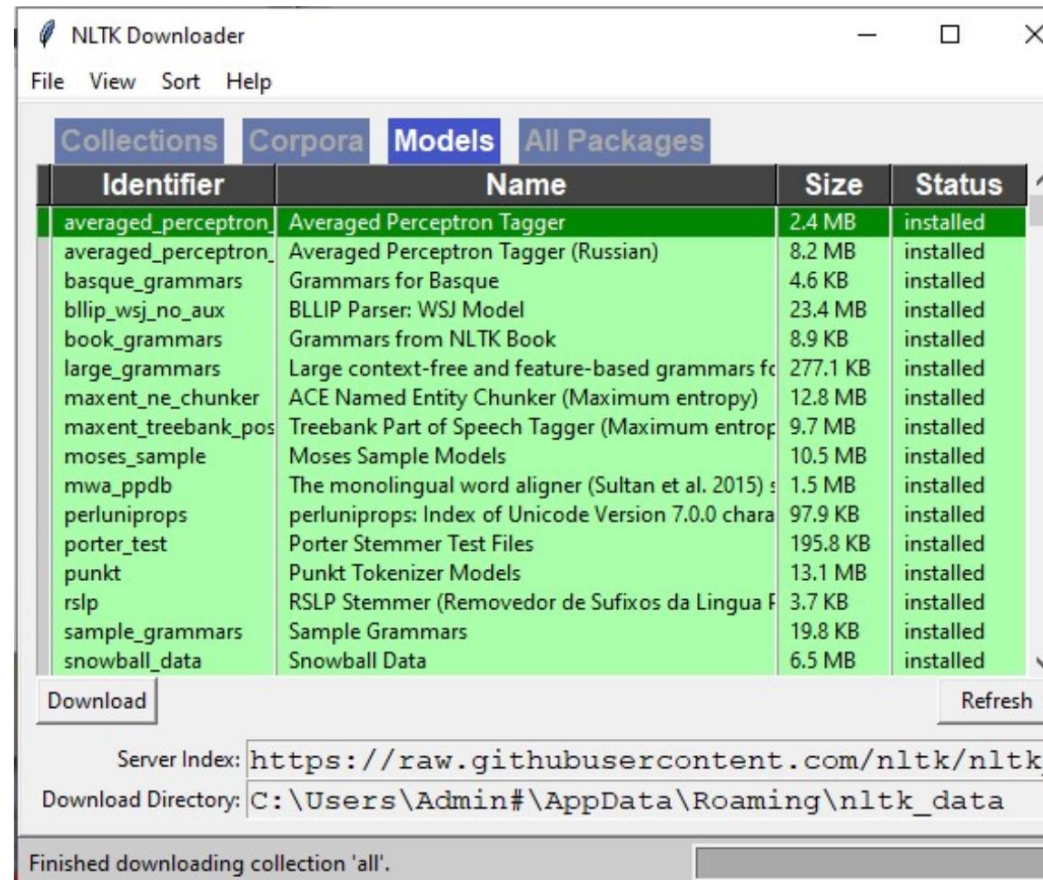
```
nltk.download()
```

This opens an interface for downloading NLTK packages and resources.



NLTK Resource Packages

The downloader can be used to install the corpora, tokenizers, taggers, grammars, and models required by NLTK.



spaCy — Industrial-Strength NLP

spaCy is an open-source library for NLP operations.

spaCy was released in 2015.

Installation:

<https://spacy.io/usage>

```
pip install spacy
```

Version check

```
import spacy
spacy.__version__
```

Quickstart

Operating system	macOS / OSX <input checked="" type="button" value="Windows"/> Linux
Package manager	<input checked="" type="button" value="pip"/> conda from source
Python version	2.x <input checked="" type="button" value="3.x"/>
Configuration	☐ virtualenv ?
Additional data	☐ Lemmatization ?
Models	☐ Chinese ☐ Danish ☐ Dutch ☒ English ☐ French ☐ German ☐ Greek ☐ Italian ☐ Japanese ☐ Lithuanian ☐ Norwegian Bokmål ☐ Polish ☐ Portuguese ☐ Romanian ☐ Spanish ☐ Multi-language

```
$ pip install -U spacy
$ python -m spacy download en_core_web_sm
```

spaCy Models and Language Packages

Installing the library alone is not enough: language models must also be downloaded.

English language package

```
python -m spacy download en
```

Small English model

```
python -m spacy download en_core_web_sm
```

Large English model

```
python -m spacy download en_core_web_lg
```

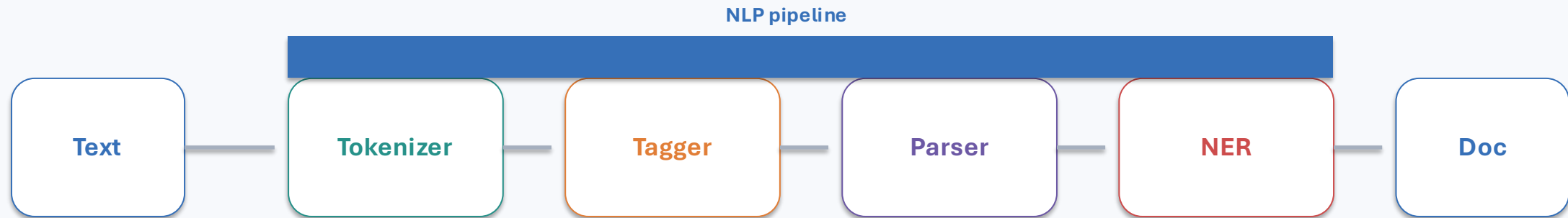
Load the model

```
nlp = spacy.load('en_core_web_sm')
```

French, German, and Spanish among available languages and notes that Arabic was not available at the time of the material.

The spaCy NLP Pipeline

a sequence in which input text passes through several NLP processing components.



Load and import the library.

Build or invoke the processing pipeline.

Represent text as tokens.

Apply tagging and inspect token attributes.

Process syntactic structure and named entities.

POS, stemming, lemmatization, and related NLP operations.

Regular Expressions — re

The Python re module provides support for Regular Expressions (regex): patterns used to search, match, and manipulate text.

Import the module

```
import re
```

re is often already available with an Anaconda installation.

Historical note

The foundations date to the 1950s and that the library emerged in the 1990s under programmer Philip Hazel.

REGEX

\b[A-Z] [a-z]+\b

Pattern Find capitalized words

Input “Natural Language Processing”

Output **Natural Language Processing**

<https://regex101.com/>

Library Ecosystem — Summary

NLTK

Classical NLP toolkit, corpora, tokenization, tagging, and educational NLP workflows.

spaCy

Production-oriented NLP pipeline with tokenization, tagging, parsing, NER, and models.

re

Pattern-based text processing through Regular Expressions.

Gensim

among the course libraries; covered later in the course.

FastText

among the course libraries; covered later in the course.

NumPy / pandas / scikit-learn / Matplotlib

General-purpose Python tools used alongside NLP workflows.